

SOOYONG KIM

Robotics & Autonomous Systems Engineer

sooyongtech.dev | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Robotics and autonomous systems engineer with experience in ROS/ROS 2, computer vision, sensor fusion, localization, path planning, control, and mobile robotics research. Strong mechanical engineering foundation in dynamics, numerical modeling, sensing, signal processing, system identification, and experimental validation. Research experience spans autonomous mobile robots at the University of Waterloo and vibration-based system identification at the University of Victoria. Currently developing and validating ROS 2-based robotics systems and perception pipelines with an emphasis on quantitative evaluation and reproducible implementation.

EDUCATION

University of Waterloo PhD Program in Mechanical & Mechatronics Engineering Voluntary Withdrawal	2020-2021
University of Victoria Master of Applied Science (MAsc), Mechanical Engineering Thesis: Predicting Regenerative Chatter in Turning Using Operational Modal Analysis	2016-2019
Pusan National University Bachelor of Science (B.Sc.), Mechanical Engineering	2005-2012

SELECTED ROBOTICS PROJECTS

Real-Time Vision Perception Pipeline

Personal Robotics Project

ROS 2 · PyTorch · Ultralytics YOLO · OpenCV · DeepSORT

- Developed a ROS 2 perception pipeline integrating YOLO-based instance segmentation, multi-object tracking, image processing, and quantitative performance logging.
- Achieved 31.95 FPS across 2,136 steady-state frames, with 18.26 ms average processing latency and 24.87 ms P95 latency on an NVIDIA RTX 5060 Laptop GPU.
- Measured 8.70 ms average inference latency and 12.52 ms P95 inference latency, processing scenes containing up to 11 detected objects per frame.
- Structured the implementation as reusable ROS 2 nodes with launch configuration, CSV-based performance reporting, Git version control, and documented experimental results.

ROS 2 EKF Sensor Fusion

Personal Robotics Project

ROS 2 Humble · Gazebo

- Implemented an Extended Kalman Filter pipeline for mobile-robot state estimation using wheel odometry and IMU measurements.
- Integrated simulated sensor streams, coordinate frames, and estimated robot states within a Gazebo-based ROS 2 environment.
- Evaluated fused-state behavior against individual sensor measurements and documented the implementation as a reproducible robotics demonstration.

LiDAR-Camera Fusion for Road Segmentation

University of Waterloo

KITTI · TensorFlow/Keras · Computer Vision

- Developed and evaluated RGB and LiDAR-camera fusion architectures for semantic road segmentation using projected Velodyne depth data and the KITTI road dataset.
- Compared RGB-only, early-fusion, and late-fusion variants of a modified SegNet architecture.
- Achieved a MaxF score of 0.9364 using late fusion, compared with 0.9010 for the RGB-only model, and evaluated accuracy, IoU, computational cost, and fusion-strategy trade-offs.

RESEARCH EXPERIENCE

University of Waterloo

May 2020 - Dec 2021

Research Assistant, Mechanical & Mechatronics Engineering

Waterloo Mechanical Systems and Control Lab

- Conducted mobile robotics research involving autonomous navigation, path planning, localization, sensing, and physical-robot experimentation.
- Developed LiDAR-camera fusion methods for semantic road segmentation using deep neural networks and the KITTI dataset.
- Implemented model-predictive-control-based mobile-robot navigation and conducted physical experiments to evaluate path-following performance.
- Supported path-planning research for indoor mobile robots in dynamic environments and physical validation using Summit mobile robots and motion-capture systems.
- Developed and validated ROS laboratory material covering SLAM, AMCL, RRT/RRT*, Gazebo, and autonomous navigation.

University of Victoria

Aug 2017 - Apr 2019

Research Assistant, Dynamics and Digital Manufacturing Lab

- Developed an Operational Modal Analysis approach for estimating machining stability from in-process vibration measurements before regenerative chatter fully developed.
- Modeled dynamic systems using state-space and time-delay formulations and performed numerical simulation, pole identification, damping estimation, and stability analysis in MATLAB.
- Designed and conducted instrumented experiments using accelerometers, National Instruments DAQ hardware, acoustic measurements, and laser Doppler vibrometry.
- Applied signal processing and system-identification methods including correlation analysis, PSD, TDPR, and modified LSCE, and validated results against physical experiments.

PROFESSIONAL EXPERIENCE

MKS Investments

2023-2026

Manager

- Reviewed and validated technical and market analyses produced by internal analysts and AI-assisted workflows, identifying incorrect assumptions, data inconsistencies, unsupported claims, and source-quality issues.
- Cross-checked technical feasibility, benchmarks, source data, and market evidence to improve the reliability of technology-development and investment decisions.
- Managed technology and market research across emerging industries and translated technical findings into decision-oriented recommendations for management and external stakeholders.

SELECTED PUBLICATION

S. Kim and K. Ahmadi, "Estimation of vibration stability in turning using operational modal analysis," *Mechanical Systems and Signal Processing*, 2019.

CORE TECHNOLOGIES

Robotics & Autonomy: ROS 2, ROS, Nav2, SLAM, AMCL, Localization, Path Planning, RRT/RRT*, Pure Pursuit, Model Predictive Control

Perception & Sensor Fusion: OpenCV, YOLO, DeepSORT, Instance Segmentation, Semantic Segmentation, LiDAR-Camera Fusion, EKF, IMU, Odometry

Programming & ML: Python, C++, MATLAB, Linux, Git, PyTorch, TensorFlow/Keras, Jupyter

Dynamics & Systems: State-Space Modeling, Numerical Analysis, System Identification, Stability Analysis, Signal Processing, FEA